

Do groups A, B, and C differ — and which pairs?

Statistical board report · $\alpha = 0.05$, two-sided · $n = 30$ per group · 1 round to finalize

Question & hypothesis

- **Omnibus** — $H_0: \mu_A = \mu_B = \mu_C$ vs. H_1 : at least one group mean differs.
- **Pairwise** (if omnibus rejects) — which of A–B, A–C, B–C differ, with family-wise error controlled.

Methods — and why these tests

The choice of test was **decided by the data, not assumed**:

- **Normality holds** for all three groups (Shapiro–Wilk p : A = 0.416, B = 0.744, C = 0.093), so a parametric route is defensible.
- **Variances are unequal** — Levene's test **$p = 0.0046$** (Bartlett agrees, $p = 0.017$), and the SDs bear it out (A = 7.77, B = 8.05, **C = 12.37**, a variance ratio ≈ 2.5).

Because homogeneity of variance fails, the **classic one-way ANOVA and Tukey HSD are invalid here** (both pool a single error variance) and are *not* reported as the result. The board used their heteroscedastic-robust replacements:

Purpose	Test used	Why
Omnibus	Welch's ANOVA	Does not assume equal variance
Omnibus backup	Kruskal–Wallis	Non-parametric cross-check
Pairwise	Welch t-tests + Holm correction	Unequal-variance-safe; controls family-wise error
A–C equivalence	TOST	"No difference" needs a positive test, not a large p
Evidence strength	JZS Bayes factors ($r = 0.707$)	Quantifies evidence, incl. <i>for</i> the null

Results

Omnibus — the groups differ. Welch's ANOVA **$F(2, 56.28) = 9.41$, $p = 0.00030$** ; Kruskal–Wallis **$H = 12.40$, $p = 0.0020$** agrees. Omnibus effect is large ($\eta^2 = 0.152$, $\omega^2 = 0.132$; Cohen's $f \approx 0.42$).

Pairwise (Holm-adjusted Welch t-tests):

Pair	Mean diff	Adj. p	Cohen's d	Bayes factor	Verdict
A vs B	–7.97	0.00075	–1.01 (large)	$BF_{10} = 100$ (extreme)	Differ
B vs C	+8.97	0.0033	0.86 (large)	$BF_{10} = 21.8$ (strong)	Differ
A vs C	–1.00	0.710	0.10 (negligible)	$BF_{01} = 3.6$ (moderate, null)	No difference detected

Conclusion

B stands apart from both A and C; A and C are statistically indistinguishable. This is *not* "all three groups differ." The two differences (A–B, B–C) are both **large and decisive** — significant after Holm correction and corroborated by extreme/strong Bayes factors — so they are not artifacts of multiplicity or the variance imbalance.

On **A vs C**, the frequentist test ($p = 0.71$) and a moderate Bayes factor ($\text{BF}_{01} = 3.6$ — the data are $\sim 3.6\times$ more likely under "no difference") both point to no effect, and the observed difference is negligible ($d = 0.10$). But this is **"no difference detected,"** not proven equivalence: a **TOST within ± 5 units fails ($p = 0.069$)**, so a small-to-moderate A–C difference cannot be ruled out.

Limitations

- **Underpowered for small effects.** At $k = 3$, $n = 30$, $\alpha = 0.05$, power is only ≈ 0.20 for a small effect ($f = 0.25$) and ≈ 0.49 even at the observed effect ($f \approx 0.42$). The design reliably catches only large effects — which is why the A–C null is "undetected," not "confirmed equal." Detecting an A–C difference the size observed would need a far larger n .
- **Bayesian A–C strength is prior-dependent.** BF_{01} for A–C moves from 2.8 (skeptical prior, $r = 0.5 \rightarrow$ only *anecdotal*) to 4.8 (wide prior, $r = 1.0$). The *direction* (favoring the null) is robust; the *strength* is borderline moderate. The A–B and B–C conclusions are prior-insensitive.
- **Independence is assumed, not verified.** The tests treat A/B/C as independent between-subjects samples. This cannot be checked from the numbers; if the data are actually paired/repeated-measures on the same units, a repeated-measures ANOVA / Friedman design would be required instead.

Sources

Every statistic above was computed and independently reproduced with the project engine, `python3 -m stat_board.engine`, against `sample_data/long.csv`. No number was hand-computed. The classic ANOVA ($F = 7.82$, $p = 0.00075$) and Tukey HSD were run but **excluded from the conclusion** because Levene's test invalidated their equal-variance assumption.

Board: 1 round · analyst $\rightarrow$ frequentist / assumptions / Bayesian critics $\rightarrow$ verifier $\rightarrow$ judge. Verifier reproduced all 8 published claims (all supported).

Appendix A — Figures

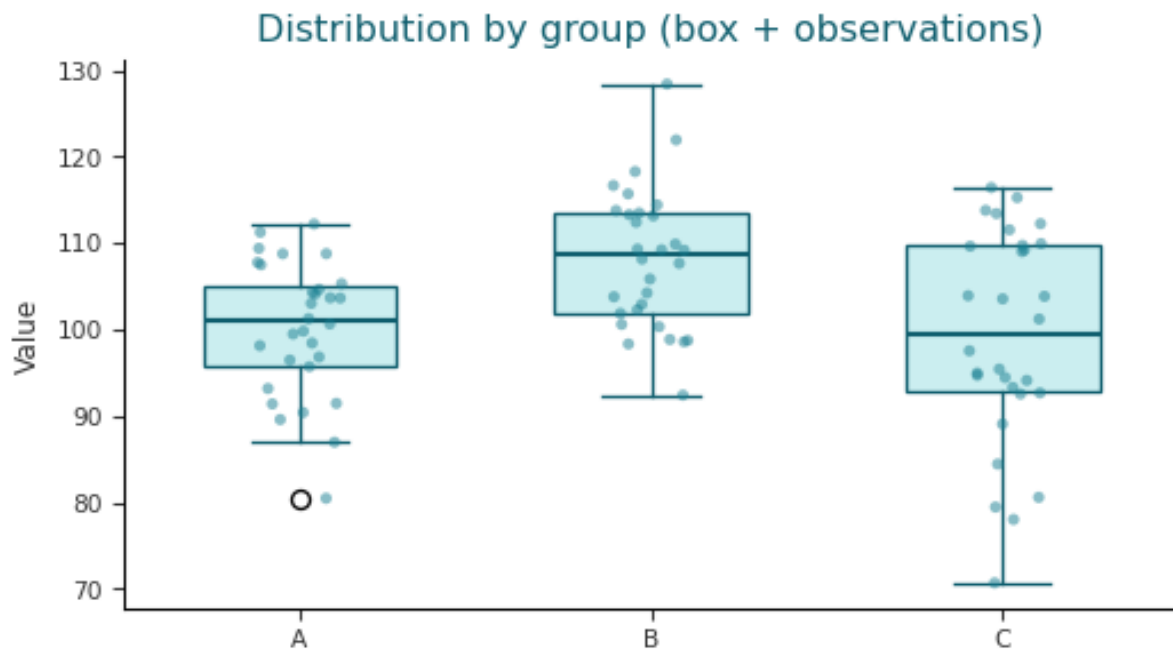


Figure 1. Box plot of each group with individual observations overlaid.

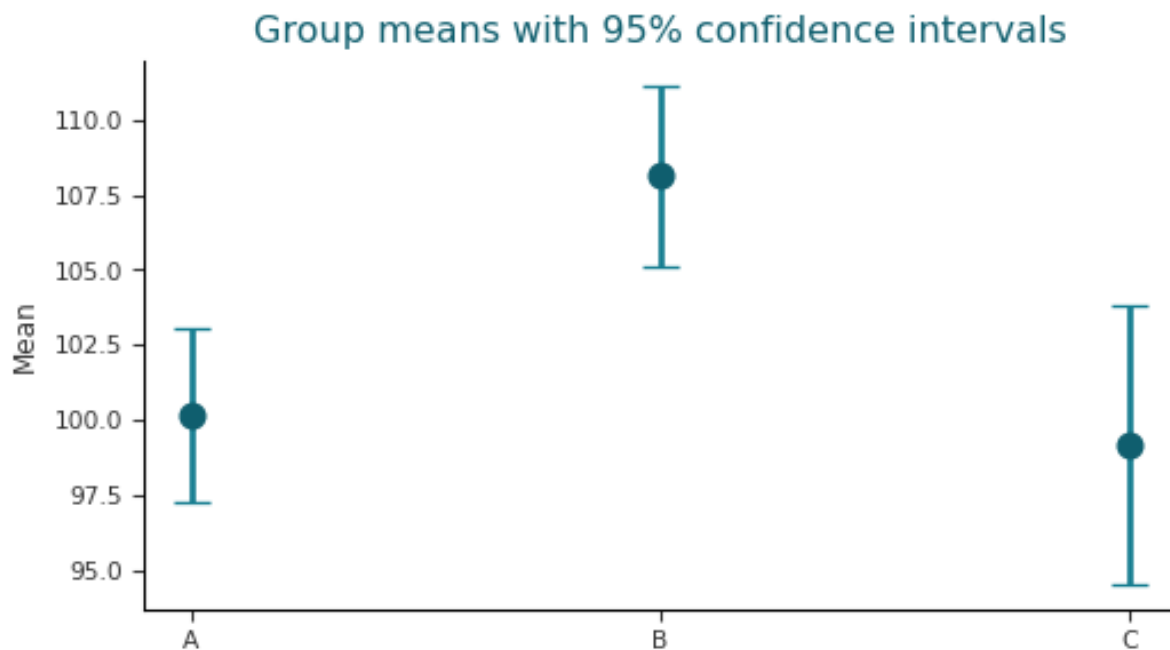


Figure 2. Group means with 95% CIs; non-overlapping bars suggest a difference.

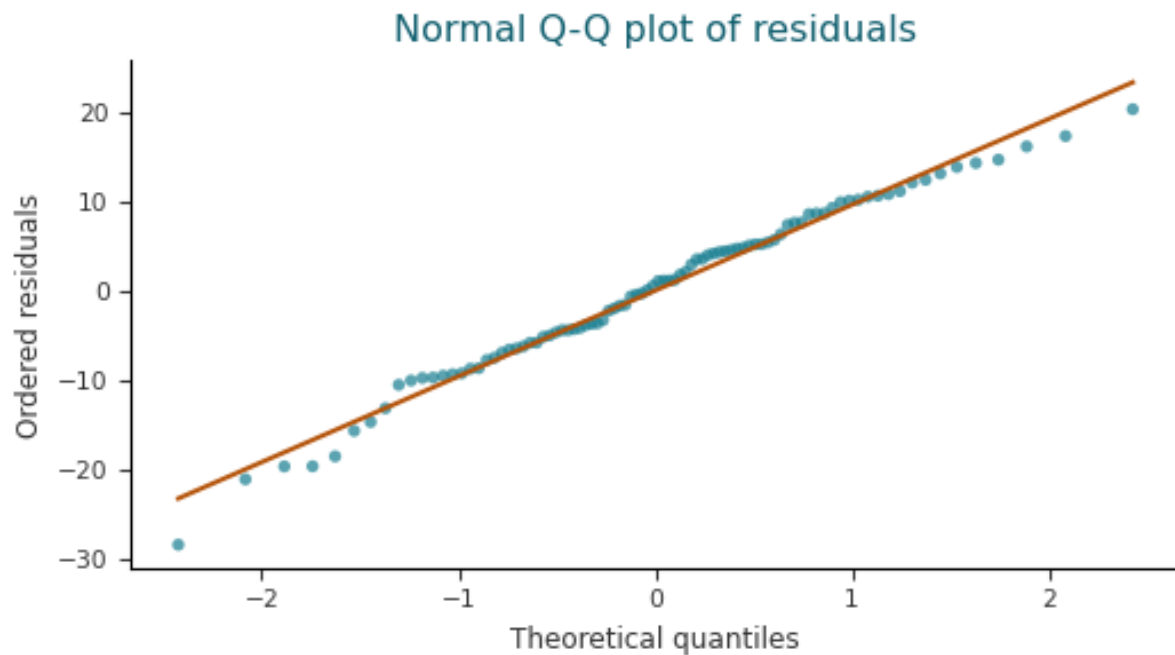


Figure 3. Normal Q-Q plot of model residuals; points on the line support the normality assumption.

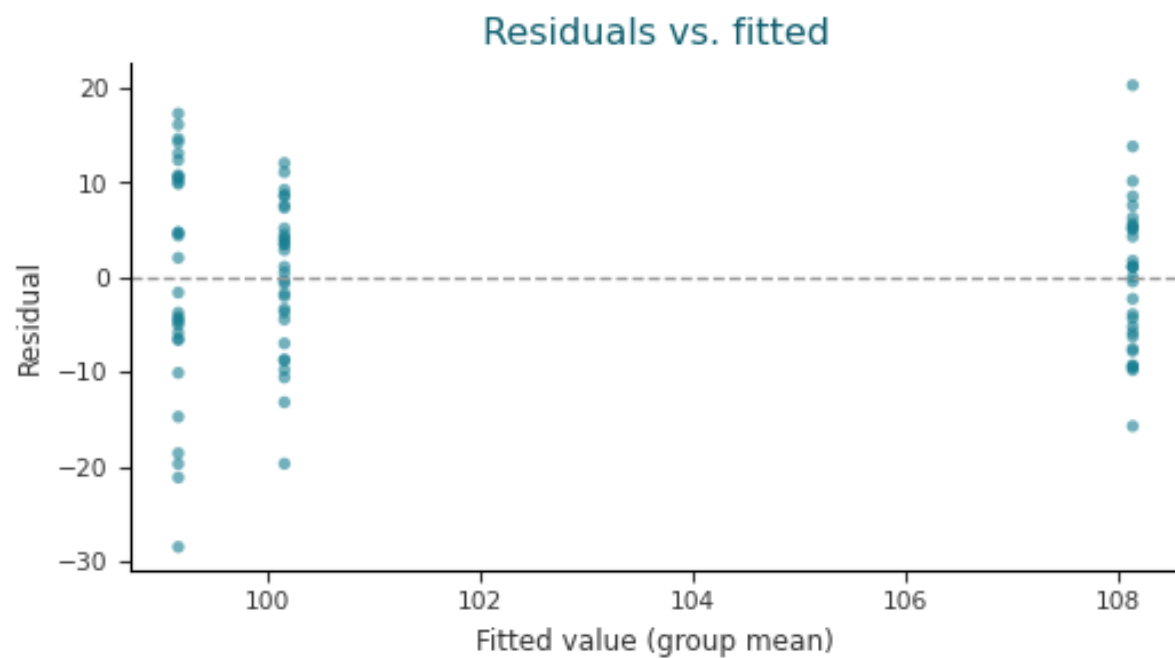


Figure 4. Residuals vs. fitted values; even vertical spread across groups supports equal variance.

Appendix B — Detailed statistics

Descriptive statistics

Group	n	Mean	SD	SEM	Median	95% CI (mean)
A	30	100.2	7.767	1.418	101	[97.27, 103.1]
B	30	108.1	8.047	1.469	108.7	[105.1, 111.1]
C	30	99.17	12.37	2.258	99.39	[94.55, 103.8]

One-way ANOVA source table

Source	SS	df	MS	F	p
Between groups	1449	2	724.5	7.818	0.0007541
Within groups	8062	87	92.67		
Total	9511	89			

Goodness of fit

R ²	η^2	ω^2	ϵ^2	Root MSE
0.1523	0.1523	0.1316	0.1329	9.626

Residual diagnostics (value – group mean)

n	Mean	SD	Min	Q1	Median	Q3	Max
90	-6.632e-15	9.518	-28.46	-6.16	0.7707	7.076	20.28

Assumption	Statistic	Verdict
Normality of residuals (Shapiro–Wilk)	W=0.9855, p=0.4184	OK
Equal variance (Levene)	W=5.721, p=0.004632	violated

Scaled effect sizes (pairwise)

Pair	Mean diff	95% CI	Cohen's d	Hedges' g	Magnitude
A vs B	-7.969	[-12.06, -3.882]	-1.008	-0.9946	large
A vs C	0.9969	[-4.361, 6.355]	0.09654	0.09529	negligible
B vs C	8.966	[3.556, 14.38]	0.8595	0.8483	large